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Flow Curvature Effects on Darrieus Turbine Blade Aerodynamics

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The effects of curvilinear flow on Darrieus turbine blade aerodynamics are described. Analysis shows that these effects can have a sizeable impact on performance for blades of large chord. Experimental data are presented which verify this forecast. Unusually large boundary-layer radial pressure gradients and virtually altered camber and incidence are identified as causal phenomena. Conformal mapping techniques are used to transform geometric airfoils in curved flow to their virtual equivalents in rectilinear flow. It is argued that flow curvature is an important determinant of Darrieus turbine blade aerodynamic efficiency and that its proper consideration will yield performance improvements, even for blades of small chord.

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Nomenclature

AR, b, C = blade aspect ratio $= b/C$, span and chord
C_D = blade drag coefficient $= D/\frac{1}{2}\rho V_R^2 Cb$
C_{D_e} = blade effective drag coefficient, Eq. (12)
$C_{D_{\min}}$ = blade minimum drag coefficient
C_L = blade lift coefficient $= L/\frac{1}{2}\rho V_R^2 Cb$
C_M = blade moment coefficient $= M/\frac{1}{2}\rho V_R^2 C^2 b$
C_p = blade power coefficient $= T_B\omega/\rho V_\infty^3 bR$
D = blade aerodynamic drag
L = blade aerodynamic lift
M = blade aerodynamic moment
p = fluid static pressure
R = turbine rotor radius
Re = Reynolds number $= \omega RC/\nu$
T_B = blade torque
TSR = blade tip speed ratio $= \omega R/V_\infty$
V_R = blade relative inflow velocity
V_∞ = freestream wind velocity
x,y = Cartesian coordinates, Fig. 2
α' = angle of attack with respect to the *local* blade relative inflow velocity
α_B = blade pitch angle from turbine radius vector
α_g = angle of attack at blade mounting point
α_i = angle of incidence resulting from flow curvature
α_v = virtual angle of attack $= \alpha_g + \alpha_i$
β = angle between V_∞ and L vectors
ϵ = inverse tip speed ratio $= V_\infty/\omega R$
θ = angle between V_∞ and R vectors
ν = kinematic viscosity
ρ = fluid mass density
ω = turbine rotor rotational speed

Introduction

ANALYTICAL and experimental research on straight-bladed Darrieus wind turbines has been conducted at West Virginia University (WVU) since 1975. These studies uncovered blade aerodynamic complexities which were beyond initial expectations. Most noteworthy are the peculiar aerodynamic phenomena associated with the orbital motion of the blades. In essence, these blades are subjected to a curvilinear flow and behave very differently than if they were immersed in a rectilinear flow. Furthermore, centrifugal forces alter the boundary layer of the airfoils rotating in this fashion. This situation presents problems in the design and analysis of all cross-flow wind turbines, because virtually all published airfoil data are derived from tests in rectilinear flow.

Recent studies[1] show that modest improvements in C_p yield desirable reductions in the cost of energy. Since these C_p improvements can be achieved by increasing blade aerodynamic efficiency, there exists ample incentive for considering the aerodynamic idiosyncrasies of rotating blades.

In the material which follows, boundary-layer centrifugal effects will be discussed first. Treatment of this subject is brief; its significance has only recently been appreciated and extensive studies of the phenomenon have not been conducted. Flow curvature effects are treated next. A simple kinematic analysis demonstrates that the turbine blade relative inflow velocity and angle of attack are unique everywhere on the chord. It is then shown how conformal mapping techniques, which transform airfoils in the curved flow field to their virtual equivalents in rectilinear flow, may be used in the aerodynamic analysis of the turbine blades. The method indicates that flow curvature effects are strongly dependent upon the blade chord to turbine radius C/R. Experimental data are introduced for two sets of blades, both of NACA 0015 airfoil section. The first set of blades had $C/R = 0.114$ and the second set had $C/R = 0.260$.

Boundary-Layer Centrifugal Effects

The importance of boundary-layer centrifugal effects has been noted in the Delft University studies[2,3] of rotating tip vanes and in the WVU flow curvature studies.[4] The existence of these centrifugal effects is explained as follows. Boundary-layer air, which is subject to viscous forces, is dragged along with the turbine blade, thereby acquiring its own rotational motion. Fluid particles in the boundary layer experience centrifugal forces which modify boundary-layer development on the blade surface. The boundary-layer air is also swept radially outward as it leaves the blade trailing edge.

The boundary-layer equations for Darrieus turbines are derived in the Appendix. These equations show that the pressure gradient (perpendicular to the surface) through the blade boundary layer is at least an order of magnitude larger than that normally experienced in curved flows. This strong

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Index categories: Wind Power; Subsonic Flow; Boundary-Layer Stability and Transition.

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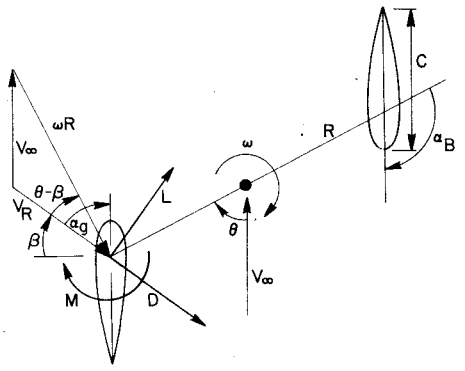


Fig. 1 Idealized turbine geometry.

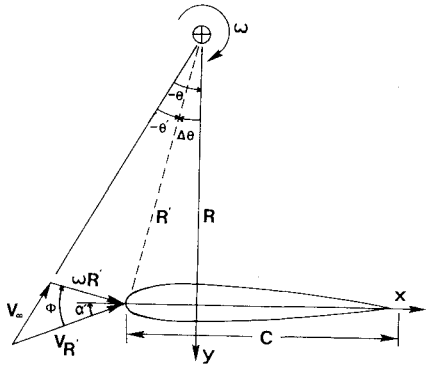


Fig. 2 Geometry for determining local velocity and angle of attack.

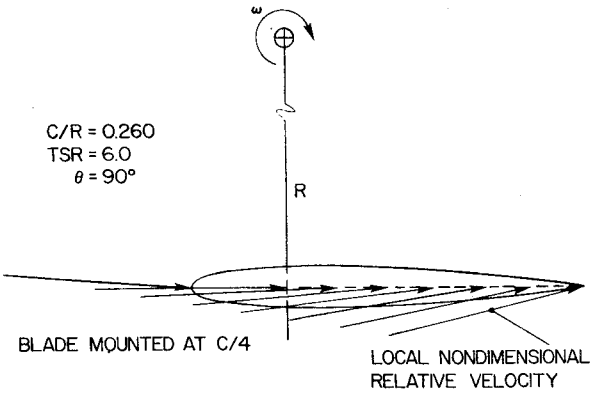


Fig. 3 Typical chordwise $V_{R'}$ and α' variation.

pressure gradient, which is proportional to $(TSR)^2$, may lead to adverse boundary-layer separation with an attendant loss of lift and increase in drag. However, the orientation of the low-pressure surface of the blade varies cyclically with orbital position because the angle of attack cyclically changes sign. Therefore, separation might be delayed on the upstream side of the turbine and accelerated on the downstream side.

Although the existence of boundary-layer centrifugal effects has been demonstrated theoretically and experimentally, their *net effect* on turbine performance is not presently known. A theoretical investigation to assess centrifugal effects on power extraction appears to be a monumental undertaking. Some simple experimental tests seem more appropriate. These should be designed specifically to determine if turbine performance degradation or enhancement is sufficiently large to warrant further study.

Flow Curvature Considerations

In the kinematic analysis of Darrieus turbines, it is common practice to define the instantaneous blade relative inflow velocity and angle of attack as those values occurring at the point of attachment of the turbine blade to its support arm. This idealized geometric representation is shown in Fig. 1. Actually, since the radial distance from the turbine axis of rotation to any point on the blade chord is unique, velocity and angle of attack are also unique at every point on the chord. From the geometry of Fig. 2, the variables θ', $\Delta\theta$, R', and ϵ' are, respectively

$$\theta' = \theta - \Delta\theta \tag{1}$$

$$\Delta\theta = -\arctan[(x/C)(C/R)] \tag{2}$$

$$R' = R/\cos\Delta\theta \tag{3}$$

$$\epsilon' = V_\infty/\omega R' = \epsilon\cos\Delta\theta \tag{4}$$

where primed symbols denote local chordwise values. The local relative inflow velocity at any point on the chord is found from the law of cosines. It is

$$V_{R'} = [(\omega R')^2 + (V_\infty)^2 - 2V_\infty\omega R'\sin\theta']^{1/2} \tag{5}$$

Equation (4) is used to simplify Eq. (5), which becomes

$$V_{R'} = \omega R'(1 + \epsilon'^2 - 2\epsilon'\sin\theta')^{1/2} \tag{6}$$

From the law of sines, the relationship for ϕ is

$$\sin\phi/V_\infty = \sin(90-\theta')/V_{R'} = \cos\theta'/V_{R'} \tag{7}$$

Using Eq. (6) for $V_{R'}$ in Eq. (7) and simplifying, we have

$$\phi = \arctan\left[\frac{\epsilon'\cos\theta'}{1-\epsilon'\sin\theta'}\right] \tag{8}$$

Finally, the local angle of attack is expressed as

$$\alpha' = \phi - \Delta\theta \tag{9}$$

Examination of these equations shows that the chordwise variation of $V_{R'}$ and α' is dependent upon the blade location relative to the freestream wind, the tip speed ratio, and the blade chord to turbine radius ratio. Of these, the last is most important; it can be shown that flow curvature effects become more pronounced as C/R increases. Figure 3 illustrates a typical chordwise variation of relative velocity and angle of attack. In this example, for a rather large $C/R = 0.260$, the α' difference from leading to trailing edge is almost 12 deg. But even for a much smaller $C/R = 0.114$ the angle-of-attack difference is approximately 5 deg. Clearly, the situation differs markedly from the normally assumed constant V_R and α_g kinematic model, and one must conclude that the aerodynamic characteristics of the airfoil in a curvilinear flowfield will not be those of the same airfoil in a rectilinear flowfield.

Although the kinematic analysis helps demonstrate flow curvature effects, it offers little insight regarding the resulting airfoil aerodynamics. However, the curvilinear flowfield of Darrieus turbines may be examined with the aid of conformal mapping techniques. By this method, the actual geometric airfoil in the curved flow may be transformed to an equivalent virtual airfoil in rectilinear flow. Local velocities and angles of attack are preserved in the process so that the transformed airfoil in an assumed rectilinear flow should exhibit the aerodynamic behavior of the geometric airfoil in orbit. Thus, if sectional airfoil data are available or can be generated for the virtual airfoil shape, the aerodynamic characteristics of the orbiting blade will be known. It should be noted that flow

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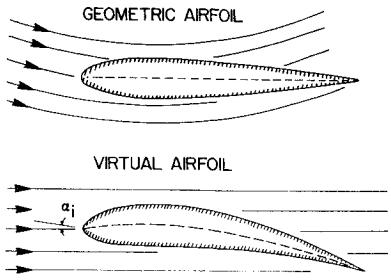


Fig. 4 Qualitative illustration of conformal transformation.

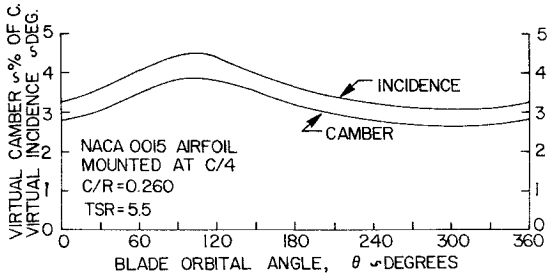


Fig. 5 Virtual camber and incidence vary with blade position θ.

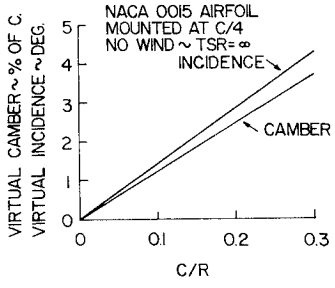


Fig. 6 Mean values of camber and incidence are dependent upon C/R.

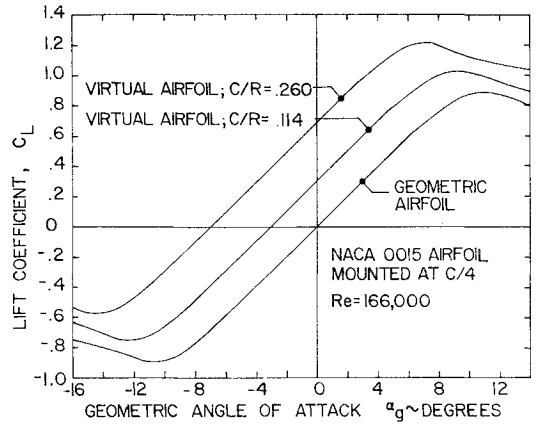


Fig. 7 Hypothetical lift curves for virtual airfoils.

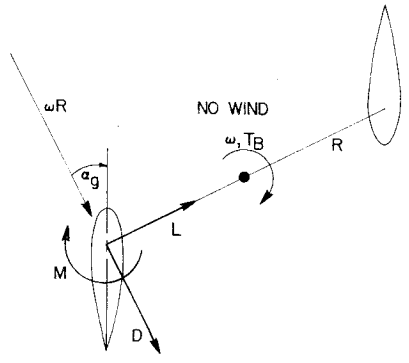


Fig. 8 Blade aerodynamic forces with no wind.

unsteadiness, blockage, blade-wake interaction, and boundary-layer centrifugal effects must still be considered.

Figure 4 qualitatively illustrates the conformal transformation procedure for a symmetrical airfoil. The transformation effects a change in camber and introduces an effective angle of incidence α_i in addition to the usual blade angle of attack. Thus, the terminology *virtual camber* and *virtual incidence* is applied to the airfoils which have a particular camber and incidence in essence, but not in fact. Camber and incidence vary with orbital position as shown in Fig. 5. However, to simplify the continuing discussion, they are assumed to be constant and equal to their mean values. The magnitude of these mean values is strongly dependent upon C/R as shown in Fig. 6.

Reference 4 may be relied upon to provide details of the virtual airfoil transformations. Equations are rigorously derived and it is shown how they may be applied to transform airfoil mean lines and surface coordinates. Procedures for calculating the airfoil maximum virtual camber and its chordwise location are presented. *Inverse* virtual airfoil transformations are also developed. They permit the investigator to describe an airfoil whose aerodynamic characteristics are desired, and to calculate the surface coordinates of the airfoil which must be fabricated in order to achieve those aerodynamic characteristics on the orbiting blade.

Virtual Airfoil Aerodynamics

The conformal transformation procedures show that symmetrical airfoils immersed in a curvilinear flow behave like cambered airfoils at incidence in a rectilinear flow. This phenomenon has important implications regarding blade aerodynamic characteristics and the performance of Darrieus turbines. These implications may be examined in part by hypothetically considering the effects of flow curvature on airfoil lift curves. The following analysis assumes an NACA 0015 airfoil, which is typical of Darrieus turbines.

Consider a blade of $C/R=0.114$ with its mounting point at the quarter chord. The transformation equations show this blade to have virtual camber of 1.4% and virtual incidence of 1.6 deg. A somewhat larger $C/R=0.260$ has virtual camber of 3.2% and virtual incidence of 3.7 deg. Examination of sectional lift data[5,6] at zero angle of attack shows an upward C_L shift of 0.10 for every 1% of camber. Therefore, the blades of $C/R=0.114$ would have $C_L=0.14$ at $\alpha_v=0$ deg and the blades of $C/R=0.260$ would have $C_L=0.32$ at $\alpha_v=0$ deg. Further modification of the lift curve results from virtual incidence, which causes an angle-of-attack shift of 1.6 deg for the blades of $C/R=0.114$ and 3.7 deg for the blades of $C/R=0.260$. These assumptions are reflected in the $C_L-\alpha_g$ curves of Fig. 7. It is of immediate interest to note the error that results if virtual camber and incidence are ignored. For example, zero lift is expected at $\alpha_g=0$ deg, whereas C_L is actually 0.30 for the smaller blades and 0.69 for the larger blades. Even more significant is the fact that the stall angle, which is $\alpha_g=11$ deg for the geometric airfoil, becomes $\alpha_g=9.4$ deg for the virtual airfoil of $C/R=0.114$ and $\alpha_g=7.3$ deg for the virtual airfoil of $C/R=0.260$. The impact of this premature blade stall is a reduction of turbine efficiency and power extraction.

Experimental evidence of virtual aerodynamics was obtained from outdoor test data. The method used is worthy of brief note. The equation for instantaneous blade torque, derived with the aid of Fig. 1, is

$$T_B(\theta)=\tfrac{1}{2}\rho V_R^2 CbR[C_L\sin(\theta-\beta)-C_D\cos(\theta-\beta)+C_M C/R] \quad (10)$$

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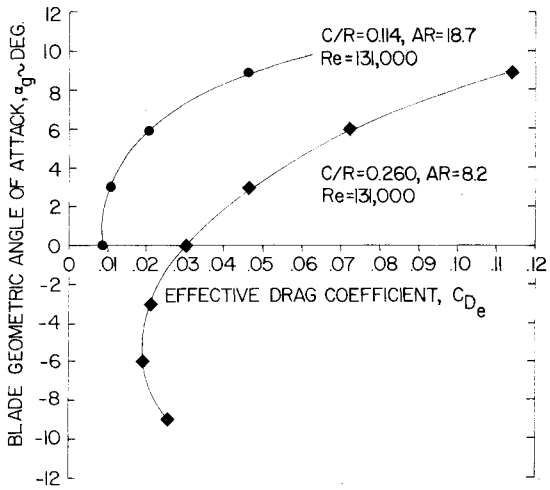


Fig. 9 Effective drag coefficients from test data.

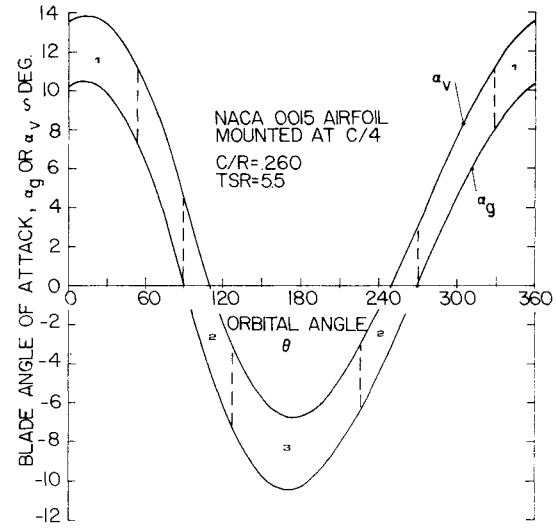


Fig. 10 Comparison of virtual and geometric angles of attack.

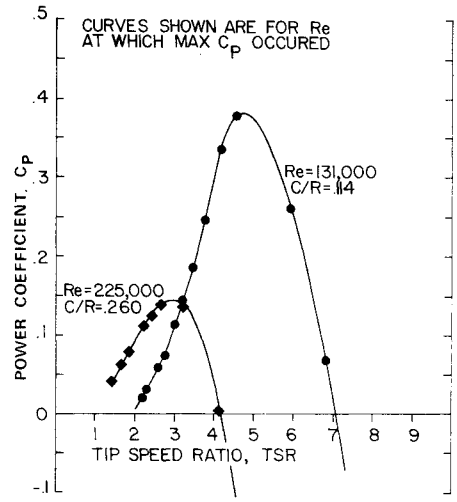


Fig. 11 Comparison of experimental C_p for $C/R=0.114$ and $C/R=0.260$.

Figure 8 shows that with no wind, neglecting induced velocities and angles of attack, the blade lift acts along the turbine radius arm and produces no torque. The drag, and moment if it exists, produce countertorque. Since ω, R, and α_g are constant, the blade torque is independent of orbital position and for a two-bladed machine it can be shown that

$$T_B=\rho\omega^2R^3Cb[C_MC/R-C_D] \tag{11}$$

An effective drag coefficient is defined which reflects the combination of aerodynamic drag and moment. This effective drag coefficient is given by

$$C_{D_e}=C_D-C_MC/R=-T_B/\rho\omega^2R^3Cb \tag{12}$$

Blades which are mounted at their quarter chords, as were the test blades, and whose airfoils behave as symmetrical sections will have $C_M=0$ and $C_{D_e}=C_D$. Furthermore, blades exhibiting symmetrical airfoil characteristics have $C_D-\alpha_g$ curves which are symmetrical about $\alpha_g=0$ deg. Thus, the existence or nonexistence of virtual aerodynamics is demonstrable by examining experimental $C_{D_e}-\alpha_g$ curves.

By testing under conditions of zero wind, or by testing at higher wind speeds and extrapolating to zero wind, the required blade torque data were generated. Curves of C_{D_e} vs α_g were developed from torque measurements taken at constant blade pitch settings and rotor rpm's. The results are plotted in Fig. 9 for one of several test *Re*. The curve for $C/R=0.260$ shows the minimum C_{D_e} occurring at $\alpha_g\neq 0$. This fact per se proves the existence of virtual camber and/or incidence.

The turbine blades, having finite span, do in fact experience induced velocities and angles of attack. Therefore, the lift vector inclination will not be as assumed, and quantitative $C_{D_e}-\alpha_g$ data are only approximate. The effect of the assumption is to exaggerate the difference in drag coefficient between blades of $C/R=0.114$ and 0.260, and to exaggerate the shift in the α_g at which minimum C_{D_e} occurs. However, the fact that the α_g shift occurs at all, regardless of its actual magnitude, is conclusive evidence of virtual aerodynamics. This is so because blades of finite span, whose airfoil sections behave as if they are symmetrical, will *always* have $C_{D_{\min}}$ at $\alpha_g=0$ deg.

Examination of $C_L-\alpha_g$ or $C_D-\alpha_g$ curves by themselves does not demonstrate the impact of flow curvature on turbine performance, since the two act in combination to produce torque. The mechanism of torque production and the manner in which it is affected by virtual aerodynamics is demonstrated as follows.

Consider blades of $C/R=0.260$ in fixed pitch operation at $TSR=5.5$ and assume the blade mounting point is the quarter chord. The conformal transformation procedures give the virtual camber and incidence as functions of orbital position as previously shown in Fig. 5. Assuming the mean orbital values of camber and incidence, the $C_L-\alpha_g$ curve of Fig. 7 applies. For this problem, with geometry as defined above, the α_v vs θ plot of Fig. 10 results. This figure provides considerable insight regarding torque production and power extraction.

Note first the regions labeled 1 in Fig. 10. Neglecting virtual incidence, the maximum angle of attack reached at $TSR=5.5$ is $\alpha_g=10.5$ deg. But virtual incidence causes an increase in this value to $\alpha_v=13.8$ deg and the stall angle is exceeded. Associated with blade stall are large drag increases which diminish torque production. So in regions 1, even though virtual camber might result in higher than expected lift coefficients, it is probable that reduced torque and C_p will be experienced. Therefore, regions 1 are areas of diminished efficiency compared to the properties expected of the geometric airfoil. Consider next the areas labeled 2 in Fig. 10. It can be shown that positive torque results from a combination of positive C_L and positive α_v on the upstream side of the turbine (90 deg $>\theta>$ 270 deg) or from the combination of negative C_L and negative α_v on the downstream side of the turbine (90 deg $<\theta<$ 270 deg). These conditions are violated in regions 2. For example, between 90 and 110 deg α_v and C_L

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are positive, and between 110 and 129 deg, even though α_v is now negative, C_L is still positive. Little power is normally extracted in these θ regions owing to low C_L at low α_v, but virtual aerodynamics make the situation much worse since countertorque results. Therefore, regions 2 are areas of diminished efficiency. Region 3 is similarly an area of diminished efficiency because $|\alpha_v|$ is much less than $|\alpha_g|$ there, and because the virtually cambered airfoils have very low C_L at negative angles of attack.

The preceding analysis shows that compensation for virtual aerodynamics is necessary in order to avoid performance degradation for large C/R. This conclusion is supported by the results of Fig. 11, which compares experimentally determined power coefficients for the blades tested on the WVU turbine. It is clear from this figure that large C/R has a detrimental effect on power extraction if flow curvature effects are ignored in blade design.

Flow Curvature and Turbine Design

Because blades of large C/R experience detrimental aerodynamic changes, one might assume that they should be avoided. This is not necessarily true. By understanding and using to advantage the mechanisms of virtual aerodynamics, large chord blades remain viable options. They may, in fact, be preferred if the desired rotor solidity can be achieved with fewer blades at lower cost. Therefore, flow curvature effects should be considered when choosing airfoil section, C/R, and blade pitch schedule.

Symmetrical airfoils have historically been chosen for Darrieus turbines because $C_L - C_D$ characteristics are symmetrical about $\alpha_g = 0$ deg, and because C_M is zero and contributes no countertorque. However, virtual camber and incidence destroy the premise upon which the choice of symmetrical airfoils is based. For a given α_g, more lift is produced upstream than downstream, and the additional α_i may even cause premature blade stall at certain values of TSR. The simplest way to alleviate this problem is to fabricate geometric airfoils whose virtual equivalents are the symmetrical airfoils originally chosen. The inverse conformal transformations[4] would be used to design the geometric airfoil for the assumed TSR, C/R, and blade mounting point. Since this can be accomplished for any C/R, the objection to large C/R is obviated. The concept may be extended to the consideration of airfoils whose virtual equivalents really are cambered. These airfoils would experience increased power extraction on the upstream side of the turbine due to increased C_L, and the diminished efficiency on the downstream side might be more than offset by the upstream improvements. This seems particularly plausible since, as a result of flow blockage, power extraction is asymmetric with respect to the cross-wind axis. At the present time, it is not certain whether net power extraction will be increased by designing for front-side or back-side optimization. The option remains to explore this possibility by judicious choice of camber and initial blade pitch setting. Such a study can be pursued only experimentally or with the aid of an accurate model of the flow blockage.

The increased complexity of electromechanical pitch variation may be justifiable for large turbines, and the potential for advantageous use of virtual aerodynamics is enhanced for these machines. For a particular configuration, the α_v vs θ relationship can be established as explained previously. It is then a simple matter to deduce the blade pitch schedule required to maintain the blade at its optimum pitch. This may require a rather large blade *flip* at $\theta = 90$ and 270 deg, where α_g normally changes sign. Again, power extraction will be asymmetric owing to both virtual camber and flow blockage. As is the case with fixed pitch operation, it is not certain whether net power extraction will be increased by optimizing the front side or the back side.

Conclusions

Analysis and experimental verification show that airfoils in curvilinear flow exhibit aerodynamic characteristics very different from what they would in rectilinear flow. Boundary-layer centrifugal effects cause normal pressure gradients which are an order of magnitude greater than expected. It is not yet possible to determine what impact this phenomenon has on Darrieus turbine performance. Flow curvature effects cause a change in the apparent camber and incidence of the rotating blades. Virtual camber influences blade aerodynamics in two important ways: It causes an upward shift in the lift curve, and it introduces an aerodynamic moment which produces countertorque. Virtual incidence influences blade aerodynamics in only one important way: It causes a lift curve shift to the left. The parameter which has the greatest impact on flow curvature effects is the ratio of blade chord to turbine radius. As C/R increases, flow curvature effects become more pronounced and performance suffers. Therefore, compensation for virtual aerodynamics is necessary in order to avoid significant performance degradation for large C/R. Similarly, compensation for virtual aerodynamics will enhance performance for small C/R. Conformal mapping techniques may be used to transform geometric airfoils in curved flow to their equivalent virtual airfoils in rectilinear flow. The inverse of this procedure is useful for designing Darrieus turbine blades.

Appendix: Boundary-Layer Considerations

Boundary-Layer Equations

The boundary-layer equations for the straight blade of a vertical axis wind turbine are derived in this section. In the development of these equations, four coordinate systems are utilized as shown in Figs. A1 and A2. They are the inertial (X, Y) frame with its origin at the center of rotation of the turbine, an auxiliary (x', y') frame with its origin attached to the pivot point of the blade and the positive x' axis in the direction of $\vec{V}_R$, a blade-fixed (x,y) frame with the positive x axis along the blade chord line, and the blade surface (ξ,η) coordinates as shown is Fig. A2. The turbine rotates with constant angular velocity ω while the blades simultaneously pitch about their pivotal axis with an angular velocity Ω. As a result of the coordinate system chosen, Ω is in general not constant.

In the blade-fixed (x,y) frame the incompressible Navier-Stokes equations and the conservation of mass equation are, respectively, $\hat{x}$ momentum:

$$\frac{Du}{Dt} = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right]$$

$$+ y\dot{\Omega} + x\Omega^2 + 2v\Omega - \omega^2 R\sin(\theta - \beta - \alpha) \qquad \text{(A1)}$$

$\hat{y}$ momentum:

$$\frac{Dv}{Dt} = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right]$$

$$- x\dot{\Omega} + y\Omega^2 - 2u\Omega + \omega^2 R\cos(\theta - \beta - \alpha) \qquad \text{(A2)}$$

and conservation of mass:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \qquad \text{(A3)}$$

where $(\dot{})$ denotes time derivatives.

Surface Coordinates

Before the boundary-layer form of the equations can be developed it is necessary to transform Eqs. (A1-A3) to surface

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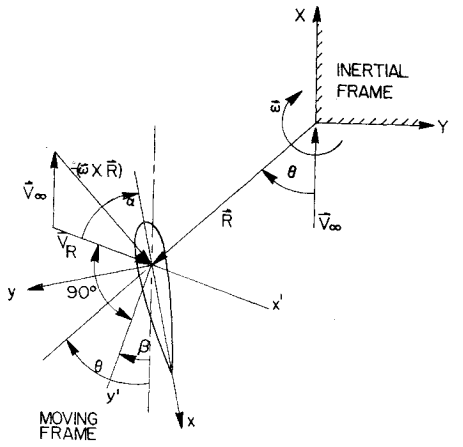


Fig. A1 Geometry for derivation of boundary-layer equations.

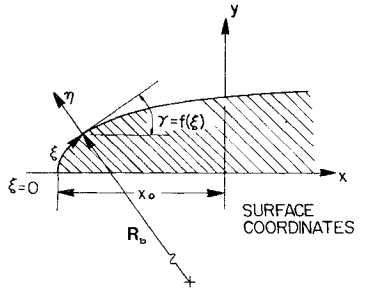


Fig. A2 Surface coordinates for boundary-layer equations.

coordinates. It can be shown that relationships between the body-fixed (x,y) Cartesian coordinates and the surface (ξ,η) coordinates are

$$x=\int_0^\xi \cos\gamma d\xi-\eta\sin\gamma-x_0 \tag{A4a}$$

$$y=\int_0^\xi \sin\gamma d\xi+\eta\cos\gamma \tag{A4b}$$

where the relationships between the corresponding unit vectors are

$$\hat{x}=\hat{\xi}\cos\gamma-\hat{\eta}\sin\gamma \tag{A5}$$

$$\hat{y}=\hat{\xi}\sin\gamma+\hat{\eta}\cos\gamma \tag{A6}$$

From the above relationships, one can readily show that the del operator in surface coordinates is related to the Cartesian operator by

$$\nabla=\hat{\xi}\frac{R_b}{R_b+\eta}\frac{\partial}{\partial\xi}+\hat{\eta}\frac{\partial}{\partial\eta} \tag{A7}$$

$$\vec{v}\cdot\nabla=v_\xi\frac{R_b}{R_b+\eta}\frac{\partial}{\partial\xi}+v_\eta\frac{\partial}{\partial\eta} \tag{A8}$$

$$\nabla^2=\frac{R_b}{R_b+\eta}\frac{\partial}{\partial\xi}\left[\frac{R_b}{R_b+\eta}\frac{\partial}{\partial\eta}\right]+\frac{R_b}{R_b+\eta}\frac{\partial}{\partial\eta}\left[\frac{R_b+\eta}{R_b}\right]\frac{\partial}{\partial\eta} \tag{A9}$$

where the body curvature R_b is a function of ξ. The relative velocity in surface coordinates is given by

$$\vec{v}=\hat{\xi}v_\xi+\hat{\eta}v_\eta \tag{A10}$$

and the substantial derivative D/Dt is given by

$$\frac{D}{Dt}=\frac{\partial}{\partial t}+v_\xi\frac{R_b}{R_b+\eta}\frac{\partial}{\partial\xi}+v_\eta\frac{\partial}{\partial\eta} \tag{A11}$$

Applying Eqs. (A7-A11) to Eqs. (A1-A3) and assuming that $R_b\gg\eta$, the transformed equations in surface coordinates are $\hat{\xi}$ momentum:

$$\frac{\partial v_\xi}{\partial t}+v_\xi\frac{\partial v_\xi}{\partial\xi}+v_\eta\frac{\partial v_\xi}{\partial\eta}=-\frac{1}{\rho}\frac{\partial p}{\partial\xi}+\nu\frac{\partial}{\partial\eta}\left[\frac{v_\xi}{R_b}+\frac{\partial v_\xi}{\partial\eta}-\frac{\partial v_\xi}{\partial\eta}\right]$$

$$+[\eta-G(\xi)]\dot{\Omega}+\Omega^2F(\xi)+2\Omega v_\eta-\omega^2R\sin(\theta-\beta-\alpha-\gamma) \tag{A12}$$

$\hat{\eta}$ momentum:

$$\frac{\partial v_\eta}{\partial t}+v_\xi\frac{\partial v_\eta}{\partial\xi}+v_\eta\frac{\partial v_\eta}{\partial\eta}=-\frac{1}{\rho}\frac{\partial p}{\partial\eta}+\nu\frac{\partial}{\partial\xi}\left[\frac{\partial v_\eta}{\partial\xi}-\frac{v_\xi}{R_b}-\frac{\partial v_\xi}{\partial\eta}\right]$$

$$-F(\xi)\dot{\Omega}-[G(\xi)-\eta]\Omega^2-2\Omega v_\xi$$

$$+\omega^2R\cos(\theta-\beta-\alpha-\gamma)+\frac{v_\xi^2}{R_b} \tag{A13}$$

and conservation of mass:

$$\frac{\partial v_\xi}{\partial\xi}+\frac{\partial v_\eta}{\partial\eta}+\frac{v_\eta}{R_b}=0 \tag{A14}$$

The exact relations for the functions $G(\xi)$ and $F(\xi)$ appearing in Eqs. (A12) and (A13) are

$$G(\xi)=\sin\gamma\int_0^\xi\cos\gamma d\xi-\cos\gamma\int_0^\xi\sin\gamma d\xi-x_0\sin\gamma \tag{A15a}$$

$$F(\xi)=\sin\gamma\int_0^\xi\sin\gamma d\xi+\cos\gamma\int_0^\xi\cos\gamma d\xi-x_0\cos\gamma \tag{A15b}$$

It is understood that $F(\xi)$ and $G(\xi)$ can be simplified for large R_b, in which case $\sin\gamma\simeq 0$ and $\cos\gamma\simeq 1$.

Boundary-Layer Form

To find the boundary-layer forms of Eqs. (A12-A14), they are first nondimensionalized. Using barred quantities to designate dimensionless variables gives

$$\bar{\Omega}=\frac{\Omega C}{V_\infty},\ \bar{\omega}=\frac{\omega C}{V_\infty},\ \bar{t}=\frac{tV_\infty}{C},\ \bar{p}=\frac{p}{\rho V_\infty^2}$$

$$\bar{R}=\frac{R}{C},\ \bar{R}_b=\frac{R_b}{C},\ \bar{F}(\xi)=\frac{F(\xi)}{C},\ \bar{G}(\xi)=\frac{G(\xi)}{C}$$

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$$\bar{\xi}=\frac{\xi}{C},\ \bar{\eta}=\frac{\eta}{C}\sqrt{Re},\ \bar{v}_\xi=\frac{v_\xi}{V_\infty},\ \bar{v}_\eta=\frac{v_\eta}{V_\infty}\sqrt{Re} \tag{A16}$$

where in this case the Reynolds number is defined as $Re=V_\infty C/\nu$.

Upon nondimensionalizing the equations and letting $Re\to\infty$, the final boundary-layer forms of the equations, written now in terms of dimensional variables, become $\hat{\xi}$ momentum ($R_b \gg \eta$):

$$\frac{\partial v_\xi}{\partial t}+v_\xi\frac{\partial v_\xi}{\partial \xi}+v_\eta\frac{\partial v_\xi}{\partial \eta}=-\frac{1}{\rho}\frac{\partial p}{\partial \xi}+\nu\frac{\partial^2 v_\xi}{\partial \eta^2}$$

$$+F(\xi)\Omega^2-G(\xi)\dot{\Omega}-\omega^2 R\sin(\theta-\beta-\alpha-\gamma) \tag{A17}$$

$\hat{\eta}$ momentum ($R_b \gg \eta$):

$$\frac{1}{\rho}\frac{\partial p}{\partial \eta}=-F(\xi)\dot{\Omega}-G(\xi)\Omega^2-2v_\xi\Omega$$

$$+\omega^2 R\cos(\theta-\beta-\alpha-\gamma)+\frac{v_\xi^2}{R_b} \tag{A18}$$

The conservation of mass equation remains unchanged from Eq. (A14).

Equations (A17), (A18), and (A14) constitute the final equations required to calculate the laminar boundary layer on a vertical axis wind turbine blade which is rotating with an angular velocity Ω with respect to the inertial frame.

Normal Pressure Gradient

The conservation of momentum equation in the direction normal to the surface ($\hat{\eta}$ momentum) as given by Eq. (A18) warrants further investigation. Classical boundary-layer analyses require that the pressure gradient normal to the blade surface be small. Usually it is of the order of the boundary-layer thickness for nonrotating systems, and for curved flows it is of the order $(\rho V_\infty^2/R_b)$.

In order to estimate the normal pressure gradient it is instructive to investigate the equations prior to writing them in surface coordinates. If the boundary-layer approximation is applied to Eq. (A2), the pressure gradient becomes

$$\frac{1}{\rho}\frac{\partial p}{\partial y}=-x\dot{\Omega}-2u\Omega+\omega^2 R\cos(\theta-\beta-\alpha) \tag{A19}$$

To gain some insight regarding the order of magnitude of this pressure gradient, it is assumed that $\dot{\Omega}=0$ and that $\epsilon\to 0$. On the basis of the second assumption, which implies a large tip speed ratio, it can be shown that $\Omega\to\omega$. With the cosine term taken to be of order unity, and since u is of order ωR, one can show from Eq. (A19) that

$$\frac{\partial p}{\partial y}\text{ is of order }\left[(TSR)^2\frac{\rho V_\infty^2}{R}\right] \tag{A20}$$

This equation shows that for a tip speed ratio of 3 or more the normal pressure gradient through the boundary layer is at least an order of magnitude larger than that resulting from curved flows.

It is of further interest to express Eq. (A20) in its nondimensional form. Using Eqs. (A16), where η is replaced by y, it can be shown that

$$\frac{\partial \bar{p}}{\partial \bar{y}}\text{ is of order }\left[\frac{(TSR)^2(C/R)}{\sqrt{Re}}\right] \tag{A21}$$

Note the appearance of the terms C/R and Re, which have been shown by other methods[3,4] to be important similarity parameters for boundary-layer centrifugal effects.

Conclusions

The boundary-layer equations for a vertical axis wind turbine blade have been derived. The classical assumption, that the pressure gradient normal to the blade surface is small, may be in serious error. In fact the pressure gradient is proportional to the square of the tip speed ratio. This implies that the usual procedure in boundary-layer theory, which is to calculate the pressures from potential theory and assume them constant through the boundary layer, may not be sufficiently accurate.

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